

Label Distribution Imputation and Bias-Corrective Representation Learning for Incomplete and Imbalanced Label Distribution

Appendix

A Supplementary Theoretical Derivations

In this section, we present a detailed proof of Theorem 1 and provide a supplementary lemma that justifies the masked KL objective under random missingness.

Theorem 1. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$R(\theta) \leq \hat{R}(\theta) + \frac{1}{\varepsilon_0} \mathcal{E}_c + \frac{L_s L_f}{\varepsilon_0} \mathcal{E}_r + C_1 \frac{\gamma}{\sqrt{N}} + C_2 \frac{r \log(2/\delta)}{N}. \quad (1)$$

Proof. Fix $(\mathbf{x}, \mathbf{d})$ and denote $\mathbf{p} = p_\theta(\mathbf{x})$ and $\mathbf{p}' = \hat{\mathbf{d}}^{\text{comp}}(\mathbf{x})$. By Lemma 1 in the main text, we have

$$|\ell_{\text{KL}}(\mathbf{d}, \mathbf{p}) - \ell_{\text{KL}}(\mathbf{d}, \mathbf{p}')| \leq \frac{1}{\varepsilon_0} \|\mathbf{p} - \mathbf{p}'\|_1. \quad (2)$$

Moreover, by the triangle inequality,

$$\|\mathbf{p} - \mathbf{p}'\|_1 \leq \|\mathbf{p} - \mathbf{d}\|_1 + \|\mathbf{p}' - \mathbf{d}\|_1. \quad (3)$$

Taking expectation over $(\mathbf{x}, \mathbf{d}) \sim P$ and using the definition of $\mathcal{E}_c$ yields the completion term $\varepsilon_0^{-1} \mathcal{E}_c$.

Next, the deviation induced by replacing the oracle representation $\mathbf{h}^\dagger(\mathbf{x}, \mathbf{d})$ with the bias-corrected representation $\mathbf{h}^*(\mathbf{x})$ is controlled by Lemma 2 in the main text:

$$\|p_\theta(\mathbf{h}^*(\mathbf{x})) - p_\theta(\mathbf{h}^\dagger(\mathbf{x}, \mathbf{d}))\|_1 \leq L_s L_f \|\mathbf{h}^*(\mathbf{x}) - \mathbf{h}^\dagger(\mathbf{x}, \mathbf{d})\|_2. \quad (4)$$

Moreover, by Lemma 1 and the assumption $p_\theta(\mathbf{x}) \in \Delta_m^{(\varepsilon_0)}$, the corresponding KL deviation is bounded by $\varepsilon_0^{-1} \|p_\theta(\mathbf{h}^*(\mathbf{x})) - p_\theta(\mathbf{h}^\dagger(\mathbf{x}, \mathbf{d}))\|_1$. Averaging over P yields the representation term $\frac{L_s L_f}{\varepsilon_0} \mathcal{E}_r$.

Finally, the remaining gap between population and empirical risks is bounded by standard concentration/complexity arguments for Lipschitz losses under imbalance-aware sampling, leading to the terms $C_1 \gamma / \sqrt{N}$ and $C_2 r \log(2/\delta) / N$ (see, e.g., [Zhang and Zhang, 2024; Zhang and Zhou, 2025; Deng *et al.*, 2024] for recent treatments). Combining the above bounds completes the proof.

Lemma 3. Assume the observation mask $\Omega_{i,j}$ is generated independently of $(\mathbf{x}_i, \mathbf{d}_i)$ with $\mathbb{P}(\Omega_{i,j} = 1) = \rho \in (0, 1]$. Let

$$\hat{R}_{\text{mask}}(\theta) = \frac{1}{N} \sum_{i=1}^N \sum_{j: \Omega_{i,j}=1} d_{i,j} \log \frac{d_{i,j}}{p_{\theta,j}(\mathbf{x}_i)}. \quad (5)$$

Then

$$\mathbb{E}_\Omega [\hat{R}_{\text{mask}}(\theta)] = \rho \hat{R}(\theta), \quad (6)$$

and hence $\rho^{-1} \hat{R}_{\text{mask}}(\theta)$ is an unbiased estimator of $\hat{R}(\theta)$. By the law of large numbers, $\rho^{-1} \hat{R}_{\text{mask}}(\theta)$ converges to $\hat{R}(\theta)$ as $N \rightarrow \infty$ under the stated missingness assumption. [Gao *et al.*, 2025; Li and Chen, 2024]

Proof. By linearity of expectation and independence of $\Omega_{i,j}$,

$$\begin{aligned} \mathbb{E}_\Omega [\hat{R}_{\text{mask}}(\theta)] &= \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^m \mathbb{E}[\Omega_{i,j}] d_{i,j} \log \frac{d_{i,j}}{p_{\theta,j}(\mathbf{x}_i)} \\ &= \frac{\rho}{N} \sum_{i=1}^N \sum_{j=1}^m d_{i,j} \log \frac{d_{i,j}}{p_{\theta,j}(\mathbf{x}_i)} = \rho \hat{R}(\theta). \end{aligned} \quad (7)$$

Therefore, $\rho^{-1} \hat{R}_{\text{mask}}(\theta)$ is unbiased for $\hat{R}(\theta)$, and consistency follows from the law of large numbers under the stated assumptions.

B Implementation Details of Compared Methods

In this section, we describe the implementation details of the models compared in this study. Unless otherwise specified, all baseline methods were implemented following their original papers or released codebases, and trained using the Adam optimizer.

DILDL. Decoupled Imbalanced Label Distribution Learning was trained using the Adam optimizer with a learning rate of 0.001, a batch size of 64, and weight decay of 1×10^{-4} .

I²LDL. Incomplete Label Distribution Learning employs iterative label completion. The model was trained with the Adam optimizer at a learning rate of 0.0005, a batch size of 128, and weight decay of 1×10^{-5} . The maximum number of label completion iterations was set to 10.

WInLDL. Weighted Incomplete Label Distribution Learning was trained with the Adam optimizer using a learning rate of 0.001, a batch size of 32, and weight decay of 1×10^{-4} . The weighting function was adapted based on missing label information.

GLDL. Graph-based Label Distribution Learning (GLDL) was optimized using the Adam optimizer with a learning rate

of 0.001 and a batch size of 64. For graph construction, the k -nearest neighbors algorithm was employed with $k = 5$. The learning rate for graph updates was also set to 0.001.

LEIC. Label Enhancement via Implicit Clustering was trained with a learning rate of 0.0001, a batch size of 64, and weight decay of 1×10^{-4} . The clustering coefficient was selected via grid search over a discrete set within the range $[0.1, 1.0]$.

LCD. Label Correlation Distribution (LCD) was trained using the Adam optimizer with a learning rate of 0.001, a batch size of 128, and weight decay of 1×10^{-5} . All hyperparameters followed the original implementation.

UALDL. Uncertainty-Aware Label Distribution Learning was trained with the Adam optimizer using a learning rate of 0.0005, a batch size of 64, and weight decay of 1×10^{-4} . The uncertainty weight for each label was computed based on prediction uncertainty.

LDIBR (Ours). LDIBR was implemented in PyTorch and trained with AdamW (learning rate 1×10^{-3} , batch size 64, weight decay 1×10^{-4}). We set the hidden dimension to $H = 128$ and fixed the BCRL basis rank to $r = 4$ with orthogonal initialization. For DALC, we used $\epsilon = 10^{-8}$ and initialized $(a, b_1, b_2) = (1.0, 1.0, 1.0)$, which were learned jointly with the network parameters. A cosine-annealing scheduler with 500-step warm-up was adopted (minimum learning rate 1×10^{-6} , $T_{\max} = 100$). The loss included the observation term and a consistency KL term (weight 1.0), with basis orthogonality (0.1) and completion smoothness (0.01) regularization; gradients were clipped to 1.0.

C Algorithm

The proposed LDIBR is detailed in Algorithm 1.

D Experimental Summary

In this section, we summarize the experimental results for LDIBR, including ablation studies on six datasets and an analysis of performance degradation under increasing missing rates.

D.1 Ablation Study

Tables 1–6 report ablation results on six benchmark datasets by removing DALC and/or BCRL from LDIBR. Disabling either module consistently degrades performance: removing DALC yields larger drops (higher KL and lower Cosine), while removing BCRL causes milder but non-negligible declines. Removing both produces the most severe deterioration, indicating that DALC and BCRL are complementary. The full LDIBR achieves the best performance across all metrics.

D.2 Further Analyses

We evaluate robustness to increasing incompleteness by varying the missing ratio from 0.4 to 0.8 and reporting KL divergence. Table 7 summarizes the training time. As shown in Fig. 1–3, KL divergence increases for all methods as the missing ratio grows, while LDIBR degrades the slowest across datasets, indicating higher robustness to label distribution sparsity.

Algorithm 1 LDIBR

Require: Dataset $\mathcal{D}$, learning rate η , batch size bs, max epochs E , basis rank r .

Ensure: Trained parameters θ (including $\mathbf{B}$).

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1: Initialize  $\theta$  (including  $\mathbf{B}$ );
2: for each epoch  $e = 1$  to  $E$  do
3:   for each mini-batch  $\{(\mathbf{x}_i, \mathbf{Y}_i, \Omega_i)\}_{i=1}^{\text{bs}} \subset \mathcal{D}$  do
4:     Stage 1: Label Distribution Completion
5:     Compute missing ratio:  $o_i = 1 - \frac{1}{m} \sum_{j=1}^m \Omega_{i,j}$ ;
6:     Generate completion parameters:
7:      $(\boldsymbol{\mu}_i, \mathbf{s}_i, \boldsymbol{\Delta}_i, \mathbf{w}_i) = \mathcal{H}([\mathbf{x}_i; \Omega_i]; \theta)$ ;
8:     Compute variability:  $\boldsymbol{\sigma}_i^2 = \exp(\mathbf{s}_i)$ ;
9:     and prior:  $\mathbf{d}_i^{\text{prior}} = \text{Softmax}(\boldsymbol{\mu}_i)$ ;
10:    Calculate gating strength:
11:     $\lambda_i = \sigma(a - b_1 o_i - b_2 \|\boldsymbol{\sigma}_i^2\|_1)$ ;
12:    Impute label distribution:
13:     $\boldsymbol{\omega}_i = \mathbf{d}_i^{\text{prior}} + \lambda_i \tanh(\boldsymbol{\Delta}_i)$ ;
14:    Fuse and normalize:
15:     $\hat{\mathbf{d}}_i^{\text{comp}} = \mathcal{N}(\mathbf{w}_i \odot \mathbf{Y}_i + (1 - \mathbf{w}_i) \odot \boldsymbol{\omega}_i)$ ;
16:    Stage 2: Representation Decomposition
17:    Decompose distribution:
18:     $\mathbf{d}_i^{(a)} = \mathbf{B} (\mathbf{B}^\top \hat{\mathbf{d}}_i^{\text{comp}})$ ;
19:    and  $\mathbf{d}_i^{(b)} = \hat{\mathbf{d}}_i^{\text{comp}} - \mathbf{d}_i^{(a)}$ ;
20:    Normalize components:
21:     $\bar{\mathbf{d}}_i^{(k)} = \mathcal{N}(\mathbf{d}_i^{(k)}), \forall k \in \{a, b\}$ ;
22:    Stage 3: Bias-Corrective Modulation
23:    Project to latent space:
24:     $\mathbf{h}_i^x = \phi(\mathbf{x}_i)$ ,
25:     $\mathbf{h}_i^{(a)} = \psi(\bar{\mathbf{d}}_i^{(a)})$ ,
26:     $\mathbf{h}_i^{(b)} = \psi(\bar{\mathbf{d}}_i^{(b)})$ ;
27:    Compute modulation parameters:
28:     $(\mathbf{g}_i^{(k)}, \boldsymbol{\beta}_i^{(k)}) = \mathcal{F}_k(\mathbf{h}_i^{(k)}), \forall k \in \{a, b\}$ ;
29:    Calculate responsibility mask:
30:     $\mathbf{m}_i^{(a)} = \sigma(\mathcal{M}([\mathbf{h}_i^x; \mathbf{h}_i^{(a)}]))$ ,
31:     $\mathbf{m}_i^{(b)} = 1 - \mathbf{m}_i^{(a)}$ ;
32:    Apply corrective modulation:
33:     $\Delta_i^{(k)} = \mathbf{m}_i^{(k)} \odot (\tanh(\mathbf{g}_i^{(k)}) \odot \mathbf{h}_i^x + \boldsymbol{\beta}_i^{(k)})$ ,
34:     $\forall k \in \{a, b\}$ ;
35:    Construct final feature:
36:     $\tilde{\mathbf{h}}_i = \mathcal{R}(\mathbf{h}_i^x + \Delta_i^{(a)} - \Delta_i^{(b)})$ ;
37:    Stage 4: Prediction and Optimization
38:    Compute predicted distribution:
39:     $\hat{\mathbf{d}}_i^{\text{p}} = \text{Softmax}(f(\mathcal{U}([\mathbf{h}_i^x; \tilde{\mathbf{h}}_i])))$ ;
40:    Calculate masked KL loss:
41:     $\mathcal{L} = \frac{1}{\text{bs}} \sum_{i=1}^{\text{bs}} \sum_{j: \Omega_{i,j}=1} Y_{i,j} \log \frac{Y_{i,j}}{\hat{d}_{i,j}^{\text{p}}}$ ;
42:    Update  $\theta$  via back-propagation:  $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$ ;
43:  end for
44: end for
45: return Trained parameters  $\theta$ .
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Method	Che.↓	Cla.↓	Can.↓	KL↓	Cos.↑	Int.↑
LDIBR (full)	0.486	2.658	7.603	0.777	0.846	0.473
w/o DALC	0.622	2.709	7.834	1.421	0.599	0.366
w/o BCRL	0.548	2.684	7.721	1.012	0.721	0.421
w/o Both	0.681	2.731	7.912	1.587	0.563	0.331

Table 1: Ablation results on *Natural Scene*.

Method	Che.↓	Cla.↓	Can.↓	KL↓	Cos.↑	Int.↑
LDIBR (full)	0.447	1.978	4.159	0.626	0.675	0.646
w/o DALC	0.583	2.024	4.323	1.034	0.512	0.546
w/o BCRL	0.511	2.006	4.245	0.781	0.603	0.589
w/o Both	0.641	2.047	4.384	1.211	0.468	0.498

Table 2: Ablation results on *SCUT-FBP*.

Method	Che.↓	Cla.↓	Can.↓	KL↓	Cos.↑	Int.↑
LDIBR (full)	0.374	1.759	3.582	0.633	0.774	0.612
w/o DALC	0.482	1.843	3.742	0.914	0.612	0.459
w/o BCRL	0.431	1.806	3.651	0.741	0.695	0.533
w/o Both	0.523	1.872	3.821	1.012	0.571	0.411

Table 3: Ablation results on *Movie*.

Method	Che.↓	Cla.↓	Can.↓	KL↓	Cos.↑	Int.↑
LDIBR (full)	0.552	2.411	6.106	1.143	0.639	0.407
w/o DALC	0.631	2.438	6.244	1.402	0.541	0.328
w/o BCRL	0.589	2.424	6.173	1.261	0.602	0.368
w/o Both	0.672	2.456	6.301	1.548	0.498	0.297

Table 4: Ablation results on *Emotion6*.

Method	Che.↓	Cla.↓	Can.↓	KL↓	Cos.↑	Int.↑
LDIBR (full)	0.570	2.481	7.279	1.779	0.661	0.421
w/o DALC	0.689	2.642	7.452	1.934	0.514	0.307
w/o BCRL	0.624	2.571	7.361	1.861	0.583	0.358
w/o Both	0.742	2.684	7.538	2.107	0.461	0.252

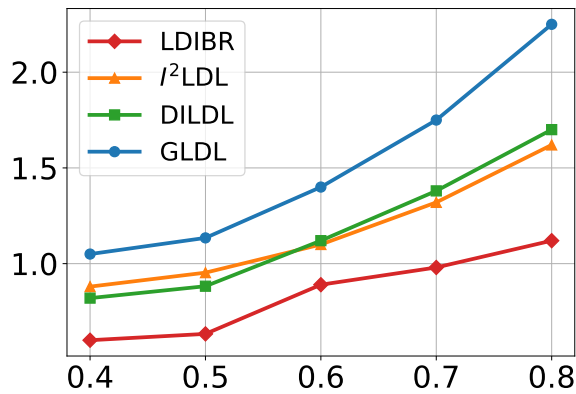
Table 5: Ablation results on *Flickr_LDL*.

Method	Che.↓	Cla.↓	Can.↓	KL↓	Cos.↑	Int.↑
LDIBR (full)	0.375	2.064	5.170	1.189	0.713	0.521
w/o DALC	0.512	2.241	5.336	1.337	0.564	0.402
w/o BCRL	0.441	2.187	5.264	1.248	0.632	0.454
w/o Both	0.588	2.298	5.417	1.462	0.503	0.319

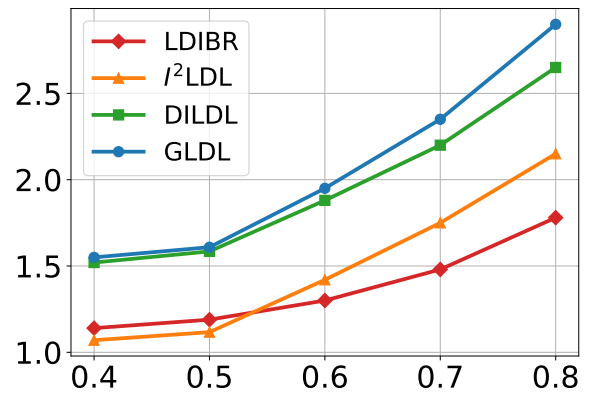
Table 6: Ablation results on *RAF-ML*.

Dataset	Flickr_LDL	Natural Scene	SCUT-FBP	Movie	Emotion6	RAF-ML
Time (s/fold)	746.13	196.27	154.58	571.62	197.62	405.83

Table 7: Average training time of LDIBR on six datasets.

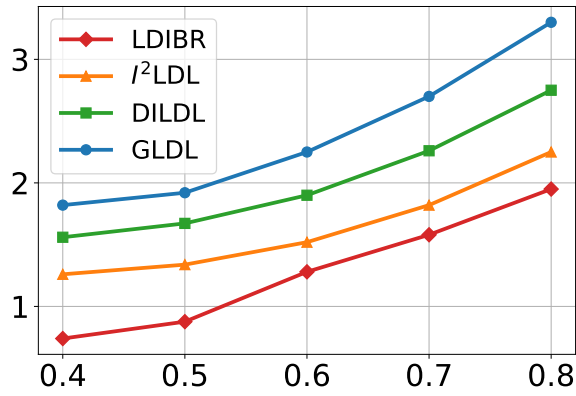


(a) *Movie*

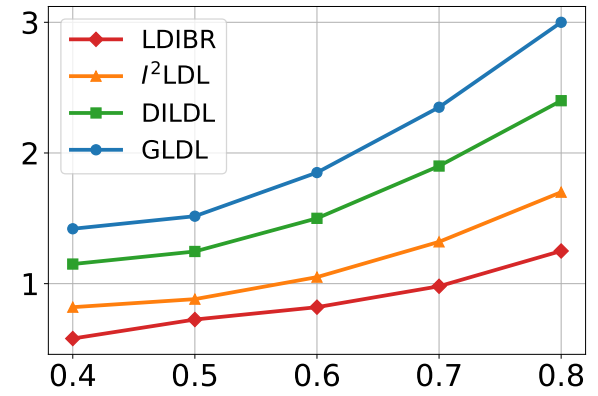


(b) *RAF-ML*

Figure 1: KL divergence under different missing ratios on different datasets. As the missing ratio increases, all methods exhibit performance degradation, while LDIBR consistently shows the slowest degradation trend.

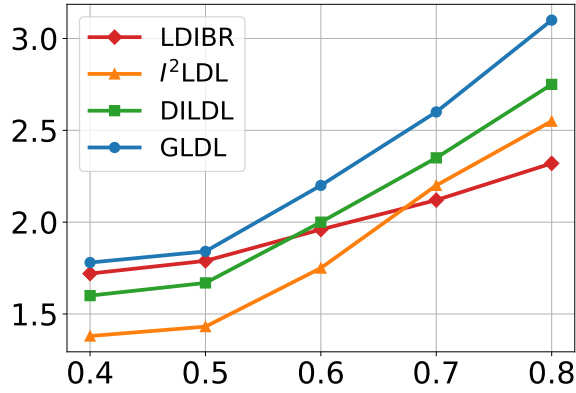


(a) *Natural Scene*

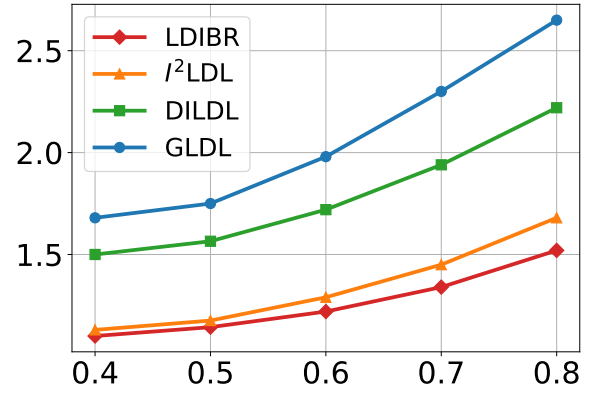


(b) *SCUT-FBP*

Figure 2: KL divergence under different missing ratios on different datasets. As the missing ratio increases, all methods exhibit performance degradation, while LDIBR consistently shows the slowest degradation trend.



(a) *Flickr-LDL*



(b) *Emotion6*

Figure 3: KL divergence under different missing ratios on different datasets. As the missing ratio increases, all methods exhibit performance degradation, while LDIBR consistently shows the slowest degradation trend.

References

- [Deng *et al.*, 2024] Xiaoge Deng, Tao Sun, Shengwei Li, Dongsheng Li, and Xicheng Lu. Stability and generalization of asynchronous sgd: Sharper bounds beyond lipschitz and smoothness. *Advances in Neural Information Processing Systems*, 37:7675–7713, 2024.
- [Gao *et al.*, 2025] Yongbiao Gao, Xiangcheng Sun, Miaogen Ling, Cao Tan, Yi Zhai, and Guohua Lv. Decoupled imbalanced label distribution learning. In *Proceedings of the International Joint Conference on Artificial Intelligence*, 2025.
- [Li and Chen, 2024] Xiang Li and Songcan Chen. No regularization is needed: efficient and effective incomplete label distribution learning. In *Proceedings of the International Joint Conference on Artificial Intelligence*, 2024.
- [Zhang and Zhang, 2024] Yi-Fan Zhang and Min-Ling Zhang. Generalization analysis for label-specific representation learning. *Advances in Neural Information Processing Systems*, 37:104904–104933, 2024.
- [Zhang and Zhou, 2025] Xianyang Zhang and Huijuan Zhou. Generalization bounds and model complexity for kolmogorov-arnold networks. In *Proceedings of the International Conference on Learning Representations*, 2025.